CO-4

Session 23 & 24

Vector Calculus

Scalar point function: A function f is said to be a scalar point function if it represents a Quantity without directions.

$$\boxed{f = f_1 + f_2 + f_3}$$

where f_1, f_2, f_3 are functions in x, y, z in coordinate plane for a point (x, y, z)

Vector point function: A function F is said to be a vector point function if it represents both Quantity & Direction.

$$\boxed{F = F_1 i + F_2 j + F_3 k}$$

where F_1, F_2, F_3 are functions in x, y, z in coordinate plane for a point (x, y, z).

$i \cdot i = 1$	$i \cdot j = 0$
$j \cdot j = 1$	$j \cdot k = 0$
$k \cdot k = 1$	$k \cdot i = 0$

Vector calculus

- Vector differentiation
  - Gradiant
  - divergance
  - Curl
- Vector Integration
  - line integrals
    - Surface integrals
    - Green's theorm
  - Stoke's theorem

Session 22 & 23

Vector differential Operator (VDO) :-

A VDO is denoted by ∇ (del)

Defined as $\boxed{\nabla = i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}}$

Gradient of a scalar point function:

The Del applying to a scalar point function is called a gradient. Denoted by grad f (or) ∇f.

where f is scalar point function.

$$\therefore \text{grad} f = \nabla f = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right) f$$

$$\boxed{\text{grad} f = \nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}}$$

Ex :- Let $f = x + 2y + 3z$

grad f = ?

$$\text{grad} f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial x} + k\frac{\partial f}{\partial x} = i(1) + j(2) + k(3)$$

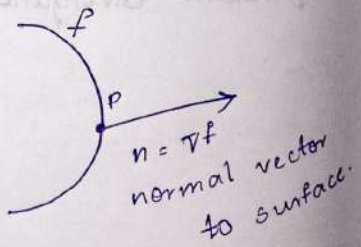
$$= \bar{i} + 2\bar{j} + 3\bar{k}.$$

$$\boxed{\text{grad} f = \bar{i} + 2\bar{j} + 3\bar{k}.}$$

Geometric Interpretition

(i) The grad f is vector normal to surface 'f'.

(ii) grad f (or) ∇f give max rate of change of f in the directions of i, j & k.



Directional derivative (DD)

Let $f(x,y,z)$ is scalar point function in the region R & p be any point in the region.

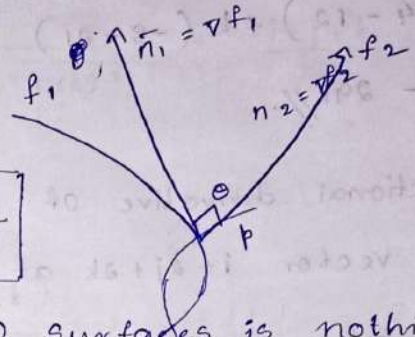
The DD of 'f' at point $P(x,y,z)$ in the direction of $\bar{a}$ is

$$\boxed{D.D = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}}$$

Angle btw 2 surfaces :

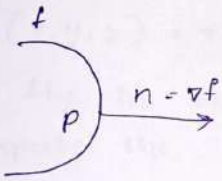
$$\cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|}$$

$$\boxed{\theta = \cos^{-1} \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|}}$$



Note : Angle between two surfaces is nothing but angle between their normals.

Unit normal vector



$$\boxed{\text{unit normal vector} = \frac{\nabla f}{|\nabla f|}}$$

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1. Find grad f, where $f = x^3 - y^3 + 3xyz$ at point $(1,-1,1)$

$$\text{grad} f = i\,3x^2 - j\,3y^2 + 3xyk(1)$$
$$= i(3x^2) - j(3y^2) + 3xyk$$
$$= i3(1) - j(3(-1)^2) + 3(1)(-1)k$$
$$= 3i - 3j - 3k$$

$$\text{grad} f = \nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$$= i(3x^2 + 3yz) - j(3y^2 + 3xz) + 3xy(1)$$
$$= i(3(1) + 3(-1)(1)) - j(3(-1)^2 + 3(1)(1)) + 3(1)(-1)k$$
$$= -3k$$

2. Find gradient of function $f = x^3y^2 - y^3z - 7(z^3 + x)$ at point $(1,2,1)$

$$\boxed{\text{grad} f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}}$$

$$= i(3x^2y^2 - 7) + j(2x^3y - 3y^2z) + k(-y^3 - 7 \times 3z^2)$$

$$= i(3(1)(2)^2 - 7) + j(2(1)(2) - 3(2)^2(1)) + k(-(2)^3 - 21(1))$$

3×4
12
-7

$$= 5i + j(4 - 12) + k(-8 - 21)$$

$$= 5i - 8j - 29k$$

3. Compute directional derivative of $f = xy^2 - y^3z - z^2x$ in the direction of vector $i + 2j + 2k$ at the point $(2, -1, 1)$

$$DD = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}$$

$$= \frac{i + 2j + 2k}{\sqrt{1 + 4 + 4}} = \frac{i + 2j + 2k}{3}$$

$$\nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$$= i(y^2 - z^2) + j(2xy - 3y^2z) + k(-y^3 - 2zx)$$

at $(2, -1, 1)$

$$= i((-1)^2 - (1)^2) + j(2(2)(-1) - 3(-1)^2(1)) + k(-(-1)^3 - 2(1)(2))$$

$$= -7j - 3k$$

$$DD = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|} = \frac{(-7j - 3k)(i + 2j + 2k)}{3}$$

$$= \frac{-14 - 6}{3} = -\frac{20}{3}$$

$$\boxed{DD = -\frac{20}{3}}$$

4. Find DD of $f = xyz$ at $(1,1,1)$ in direction of $i+j+k$

$$DD = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}$$

$$\nabla f = i\frac{\partial f}{\partial x} + j\frac{\partial f}{\partial y} + k\frac{\partial f}{\partial z}$$

$$= i(yz) + j(xz) + k(xy)$$

$$= i + j + k.$$

$$\frac{\bar{a}}{|\bar{a}|} = \frac{i+j+k}{\sqrt{(1)^2+(1)^2+(1)^2}} = \frac{i+j+k}{\sqrt{3}}$$

$$DD = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}$$

$$= \frac{(i+j+k)(i+j+k)}{\sqrt{3}} = \frac{1+1+1}{\sqrt{3}} = \frac{3}{\sqrt{3}} = \sqrt{3}$$

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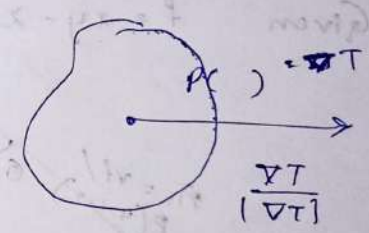
5. The temp at point (x, y, z) in the space is given by $T(x,y,z) = x^2+y^2+z$. A mosquito located at $(-1,+1,2)$ desires to fly in such a direction that it gets cooled faster compute the direction in which it should fly.

Sol The given temp. function

$$T(x,y,z) = x^2+y^2+z^2$$



the gradient of temperature is given by $\text{grad}\,T = \nabla T$

$$\nabla T = i\frac{\partial T}{\partial x} + j\frac{\partial T}{\partial y} + k\frac{\partial T}{\partial z}$$

$$\text{grad}\,T = i(2x) + j(2y) + k(1).$$

$$\text{grad}\,T = 2x\bar{i} + 2y\bar{j} + \bar{k}$$

at $(-1,+1,2)$ $= -2(1)i + 2(+1)j + k$

$$= -2i + 2j + k$$

The unit normal gradient vector $= \frac{\nabla T}{|\nabla T|} = \frac{-2i+2j+k}{3}$

To get cooled fast, it should travel in direction of $\frac{\nabla T}{|\nabla T|}$

$$= \frac{-2i+2j+k}{3}$$

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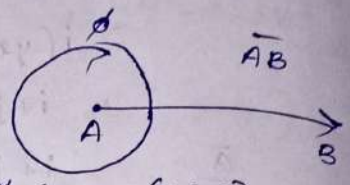
5. Find the directional derivative of $\phi = xy - yz - zx$ at A, in the direction of AB, where $A = (1,2,0)$, $B = (1,0,3)$.

Sol Given scalar point function.

$$\phi = xy - yz - zx$$

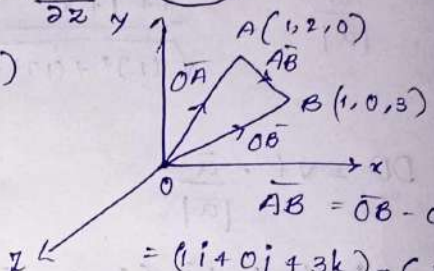


The gradient of $\phi \Rightarrow \nabla\phi = i\frac{\partial\phi}{\partial x} + j\frac{\partial\phi}{\partial y} + k\frac{\partial\phi}{\partial z}$

$$= \nabla\phi = i(y-z) + j(x-z) + k(-y-x)$$

at a point $A(1,2,0)$

$$\nabla\phi = 2\bar{i} + \bar{j} - 3\bar{k}$$



$$\bar{AB} = \bar{OB} - \bar{OA}$$

$$= (1i + 0j + 3k) - (i + 2j + 0k)$$

$$\bar{AB} = -2j + 3k$$

$$D.D = \nabla\phi \cdot \frac{\bar{AB}}{|\bar{AB}|}$$

$$\boxed{D.D = \nabla\phi \cdot \frac{\bar{AB}}{|\bar{AB}|}}$$

$$= \frac{(2i + j - 3k)\cdot(-2j + 3k)}{\sqrt{4+9}}$$

$$= \frac{-2-9}{\sqrt{13}} = -11/\sqrt{13}$$

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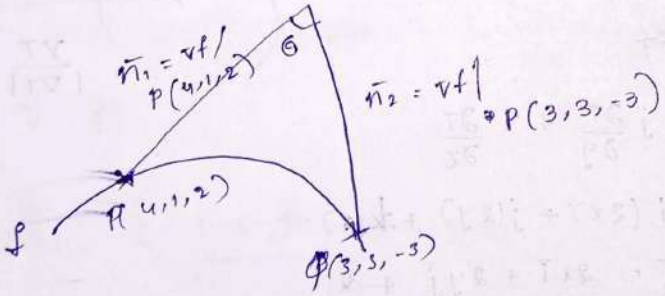
6. Identify the angle between the normal to the surface $xy = z^2$ at points $(4,1,2)$ & $(3,3,-3)$

Sol Given $f = xy - z^2$.

$xy - z^2 = 0$

$yi + xj - 2zk = 0$



The normal vector of surface f, at point P(4,1,2) is

$$\bar{n}_1 = \nabla f|_{\text{at } P(4,1,2)} = [\bar{i}y + \bar{j}x - 2z\bar{k}] = \bar{i} + 4j - 4\bar{k}.$$

The second normal vector of f_2 at point Q (3,3,-3) is

$$\bar{n}_2 = \nabla f\Big|_{\text{at } Q(3,3,-3)} = \bar{i}y + \bar{j}x - 2z\bar{k} = 3\bar{i} + 3\bar{j} + 6\bar{k}$$

Let angle between normal to surface is 'θ'.

$$\cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|} = \frac{(\bar{i} + 4\bar{j} - 4\bar{k})(3\bar{i} + 3\bar{j} + 6\bar{k})}{\sqrt{1+16+16}\sqrt{9+9+36}}$$

$$= \frac{3+12-24}{\sqrt{33}\sqrt{54}} = \frac{-9}{\sqrt{11}\sqrt{3}\sqrt{9}\sqrt{6}}$$

$$= \frac{-\cancel{9}}{\sqrt{11}\sqrt{3}\cdot\cancel{3}\sqrt{3}\sqrt{2}} = \frac{-1}{\sqrt{22}}$$

$$\theta = \cos^{-1}\left(\frac{-1}{\sqrt{22}}\right) \Rightarrow \boxed{\theta = \cos^{-1}\frac{1}{\sqrt{22}}}$$

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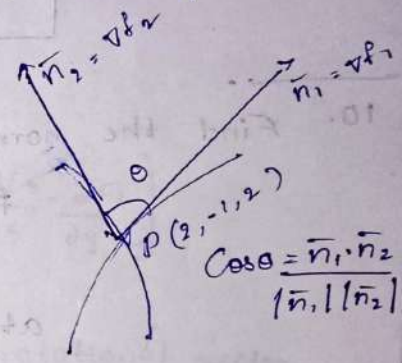
7. Determine the angle between surfaces $x^2+y^2+z^2=9$ & $x^2+y^2-3=z$ at (2,-1,2)



Given $f_1 = x^2+y^2+z^2-9$

$f_2 = x^2+y^2-z-3$

The normal vector of surface f_1, at (2,-1,2)

$$\bar{n}_1 = \nabla f_1\Big|_{\text{at } P(2,-1,2)}$$

$$\bar{n}_1 = 2x\bar{i} + 2y\bar{j} + 2z\bar{k}$$

at (2,-1,2) $= 4i - 2j + 4k$

$$\bar{n}_2 = 2x\bar{i} + 2y\bar{j} - z\bar{k}$$

at (2,-1,2) $= 4i - 2j - k$

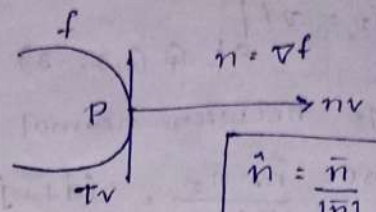
Let θ be angle between 2 surfaces is angle between their normals.

$$\cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|} = \frac{(4i - 2j + 4k)\cdot(4i - 2j - k)}{\sqrt{16+4+16}\sqrt{16+4+1}}$$

$$= \frac{16+4-4}{\sqrt{36}\sqrt{21}} = \frac{\cancel{16}\,8}{\cancel{6}_{3}\sqrt{21}}$$

$$\boxed{\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right)}$$

Unit normal vector to Surface



$$\boxed{\hat{n} = \frac{\bar{n}}{|\bar{n}|} = \frac{\nabla f}{|\nabla f|}}$$

9. Find unit normal vector to the surface $z^2 = 4(x^2+y^2)$ at point $(1,0,2)$

Sol Given surface $f = 4x^2 + 4y^2 - z^2$

$$\nabla f = 8xi + 8yj - 2zk$$

at $(1,0,2)$ = $\nabla f = 8i - 4k$.

unit normal vector of given surface f, at point

$$\boxed{\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{8i - 4k}{\sqrt{80}}}$$

10. Find the normal vector to surface $x^2+y^2-z^2=1$ at $(1,3,3)$

$$f = x^2 + y^2 - z^2 - 1$$

$$\nabla f = 2xi + 2yj - 2zk$$

at $(1,3,3)$ = $2i + 6j - 6k$

$$\hat{n} = \frac{\nabla f}{|\nabla f|} = \frac{2i + 6j - 6k}{\sqrt{4+36+36}}$$

$$\hat{n} = \frac{2i + 6j - 6k}{\sqrt{76}}$$

Session - 24

Divergence & curl of a given vector point function.

Divergence of a vector point function :-

Let $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$ be a vector, continuously differentiable on every point in 3-D plane, then the divergence of $\bar{F}$, is denoted by div$\bar{F}$ (or) $\nabla \cdot \bar{F}$

Defined as $\operatorname{div} \bar{F} = \nabla \cdot \bar{F} = \left(i\frac{\partial}{\partial x} + j\frac{\partial}{\partial y} + k\frac{\partial}{\partial z}\right) \cdot (F_1 \bar{i} + F_2 \bar{j} + F_3 \bar{k})$

$$\boxed{\operatorname{div} \cdot \bar{F} = \nabla \cdot \bar{F} = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}}$$

Solenoidal vector :-

A vector $\bar{F} = F_1 \bar{i} + F_2 \bar{j} + F_3 \bar{k}$ is said to be solenoidal vector, is $\boxed{\operatorname{div} \bar{F} = 0}$ (or) $\boxed{\nabla \cdot \bar{F} = 0}$

Curl of a vector point function:

Let $\bar{F} = F_1 \bar{i} + F_2 \bar{j} + F_3 \bar{k}$ be a vector, continuously differentiable on every point in 3-D plane, then the curl of $\bar{F}$, is denoted curl $\bar{F}$ (or) $\nabla \times \bar{F}$.

Denoted as $\operatorname{curl} \bar{F} = \nabla \times \bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{vmatrix}$

$$= \bar{i}\left[\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z}\right] - \bar{j}\left[\frac{\partial F_3}{\partial x} - \frac{\partial F_1}{\partial z}\right] + \bar{k}\left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y}\right)$$

Irrotational vector

A vector $\bar{F} = F_1 \bar{i} + F_2 \bar{j} + F_3 \bar{k}$ is said to be irrotational vector by $\boxed{\operatorname{Curl} \bar{F} = \bar{0}}$

$$\boxed{\bar{0} = 0\bar{i} + 0\bar{j} + 0\bar{k}}$$

---

1. If $\bar{f} = x^2 y \bar{i} - 2y^2 z \bar{j} + 3z^2 x \bar{k}$ find $\operatorname{div} \bar{f}$ at $(0, -1, 1)$

Sol Given $\bar{f} = x^2 y \bar{i} - 2y^2 z \bar{j} + 3z^2 x \bar{k}$

where $\boxed{\bar{f} = f_1 i + f_2 j + f_3 k}$

$f_1 = x^2 y$, $f_2 = -2y^2 z$, $f_3 = 3z^2 x$.

$$\operatorname{div} \bar{f} = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z} = 2xy + (-4yz) + 6zx.$$

at $(0,-1,1)$ $\quad div\,\bar{f} = -4(-1)(1) = 4$

$$\boxed{div\,\bar{f} = 4}$$

---

2. Determine 'p' if $\bar{F} = (x+3y)\bar{i} + (y-2z)\bar{j} + (x+pz)\bar{k}$ is solenoidal vector.

Given $\bar{F} = (x+3y)\bar{i} + (y-2z)\bar{j} + (x+pz)\bar{k}$

where $\cancel{\bar{F} = (x+3y)\bar{i} + y}$

$$\boxed{\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}}$$

Since, F is solenoidal vector, $div\,\bar{F} = 0$.

$$\bar{\nabla}\cdot\bar{F} = 0$$

$$\frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z} = 0$$

$$1 + 1 + p = 0$$

$$\boxed{p = -2}$$

---

3. If $\bar{f} = -x\bar{i} - y\bar{j} - z\bar{k}$, calculate curl $\bar{f}$.

Given $\bar{f} = -x\bar{i} - y\bar{j} - z\bar{k}$

$$curl\,\bar{f} = \bar{\nabla}\times f = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ -x & -y & -z \end{vmatrix}$$

$$curl\,\bar{f} = i[0-0] - j[0-0] + k[0-0]$$

$$= 0\bar{i} + \bar{j} + 0\bar{k} = \bar{0}$$

$$\boxed{curl\,\bar{f} = \bar{0}}$$

---

4. Find divergence & curl of $\bar{v} = (xyz)i - (2x^2y)j - 3y^2z\bar{k}$ at $(2,-1,3)$

Sol $\quad \bar{v} = (xyz)\bar{i} - (2x^2y)\bar{j} - 3y^2z\bar{k}$

$$\boxed{\bar{v} = v_1\bar{i} + v_2\bar{j} + v_3\bar{k}}$$

diverage of $\bar{v}$ is div $\bar{v}$ (or) $\nabla \cdot \bar{v}$

$$\text{div } \bar{v} = \frac{\partial v_1}{\partial x} + \frac{\partial v_2}{\partial y} + \frac{\partial v_3}{\partial z}$$

$$\text{div } \bar{v} = yz - 2x^2 - 3y^2$$

at $(2,-1,3)$ $\quad \text{div } \bar{v} = (-1)(3) - 2(2)^2 - 3(-1)^2$

$$= -3 - 8 - 3 = -14$$

$$\boxed{\text{div } \bar{v} = -14}$$

$$\text{curl } \bar{v} = \nabla \times \bar{v} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xyz & -2x^2y & -3y^2z \end{vmatrix}$$

$$\text{curl } \bar{v} = i[-6yz - 0] - j[0 - xy] + k[-4xy - xz]$$

at $(2,-1,3)$ $\quad \bar{v} = 18\bar{i} - 2\bar{j} + 2\bar{k}$.

5. show that the fluid motion

$\bar{f} = (y+z)\bar{i} + (x+z)\bar{j} + (x+y)\bar{k}$ is irrotational.

Sol $\quad \text{curl } \bar{f} = \nabla \times \bar{f} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (y+z) & (x+z) & (x+y) \end{vmatrix}$

$$\text{curl } f = i[1-1] - j[1-1] + k[1-1]$$

$$= 0\bar{i} + 0\bar{j} + 0\bar{k} = \bar{0}$$

$$\boxed{\text{curl } \bar{f} = \bar{0}}$$

$\therefore \bar{f}$ is irrotational vector.

6. Determine a,b,c if $\bar{f} = (2x+3y+az)i + (bx+2y+3z)j + (2x+cy+3z)k$ is irrotational

Sol :- Given vector $\bar{f}$ is irrotational vector,

$$\text{curl } \bar{f} = 0$$

$$\nabla \times \bar{f} = \bar{0}.$$

$$\Rightarrow \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2x+3y+az & bx+2y+3z & 2x+cy+3z \end{vmatrix} = 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$$= i[c-3] - j[2-a] + k[b-3] = 0\bar{i} + 0\bar{j} + 0\bar{k}$$

$c - 3 = 0$

$\boxed{c = 3}$ $\boxed{a = 2}$ $\boxed{b = 3}$

Vector Integration

(i) Line Integral

(ii) Green's theorem for plane

(iii) Surface integrals

(iv) Stoke's theorem.

Session - 25

Let $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$ be any vector point function.

Position vector $\bar{r} = x\bar{i} + y\bar{j} + z\bar{k}$.

The line integral of $\bar{F}$ over the curve 'C' moves from point 'A' to point 'B' is given

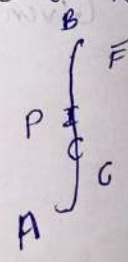
$$\boxed{I = \oint_C \bar{F} \cdot d\bar{r}}$$ $d\bar{r}$ is displacement vector.

$$\Rightarrow \boxed{I = \oint_C (F_1 dx + F_2 dy + F_3 dz)}$$

Workdone by Force Function $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$:

Workdone by the force function $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$, for moving a particle 'P' from point 'A' to point 'B', along circle 'C' is

$$\boxed{W = \oint_C \bar{F} \cdot d\bar{r}}$$



2. Compute workdone by the force function $\bar{F} = 2xy\bar{i} - 3z\bar{j} + 5x\bar{k}$, when it moves a particle along curve $x = t^2+1$, $y = 2t^2$, $z = t^3$ from $t = 1$ to 2.

Sol force function,

$$\bar{F} = 2xy\,\bar{i} - 3z\,\bar{j} + 5x\,\bar{k}$$

The work done by Force $\bar{F}$, to move particle from point A to point 'B', along curve 'C' is

$x = t^2+1$
$y = 2t^2$
$z = t^3$
$t \rightarrow$ varies from 1 to 2

$$w = \oint_C \bar{F}\cdot d\bar{r}$$

$$= \oint_C (2xy\,\bar{i} - 3z\,\bar{j} + 5x\,\bar{k})\cdot(i\,dx + j\,dy + k\,dz)$$

$$= \oint_C 2xy\,dx - 3z\,dy + 5x\,dz.$$

where 'C' is curve given by

$x = t^2+1 \Rightarrow dx = 2t\,dt$

$y = 2t^2 \Rightarrow dy = 4t\,dt$

$z = t^3 \Rightarrow 3t^2\,dt$

$$w = \oint_C 2(t^2+1)(2t^2)\,2t\,dt - 3(t^3)\,4t\,dt + 5(t^2+1)\,3t^2\,dt$$

$$= \oint_C 8t^3(t^2+1) - 12t^4 + 15t^2(t^2+1)]\,dt$$

$$= \oint_C 8t^5 + 8t^3 - 12t^4 + 15t^4 + 15t^2\,dt$$

$$t = \left[\frac{8t^6}{6} + \frac{3t^5}{5} + \frac{8t^4}{4} + \frac{15t^3}{3}\right]_1^2$$

$$t = \left[\frac{8(2)^6}{6} + \frac{3}{5}(2)^5 + \frac{8(2)^4}{4} + \frac{15}{3}(2)^3\right] - \left[\frac{8(1)}{6} + \frac{3(1)}{5} + \frac{8(1)}{4} + \frac{15}{3}\right]$$

$$t = \left[\frac{8 \times \cancel{64}^{32}}{\cancel{6}_3} + \frac{3 \times 32}{5} + \frac{\cancel{8}^{2} \times 16}{\cancel{4}} + \frac{\cancel{15}^{5}}{\cancel{3}} \times 8\right] - \left[\cancel{8}^{4}/\cancel{6}_3 + 3/5 + \cancel{8}^{2}/\cancel{4} + \cancel{15}^{5}/\cancel{3}\right]$$

$$= 167 //.$$

---

3. If $\bar{F} = 3xy\,i - y^2 j$, evaluate $\oint_C \bar{F}\cdot d\bar{r}$ where C is curve $y = 2x^2$ in the XY-plane from (0,0) to (1,2)

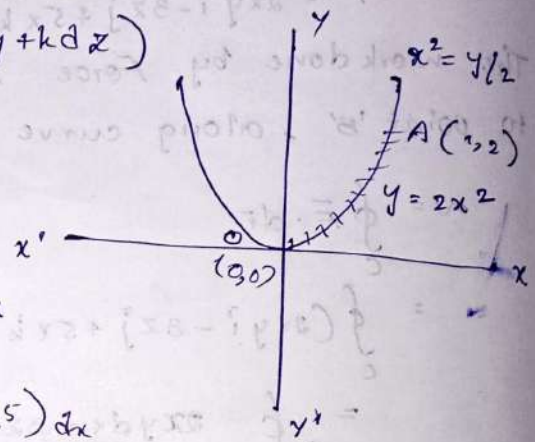
Given $\bar{F} = 3xy\,i - y^2 j$

now, $\oint_C \bar{F}\cdot d\bar{r} = \oint_C (3xy\,i - y^2 j)\cdot(i\,dx + j\,dy + k\,dz)$

$$= \oint_C 3xy\,dx - y^2\,dy$$

$\Rightarrow dy = 4x\,dx$

$$\Rightarrow \oint_C 3x(2x^2)\,dx - (2x^2)^2\, 4x\,dx$$



$$\Rightarrow \oint_C \bar{F}\cdot d\bar{r} = \int_0^1 (6x^3 - 16x^5)\,dx$$

$$= \left[\frac{6x^4}{4} - \frac{16x^6}{6}\right]_0^1$$

$$= \frac{\cancel{6}^{3}}{\cancel{4}_2} - \frac{\cancel{16}^{8}}{\cancel{6}_3}$$

$$= \frac{9 - 16}{6} = -7/6.$$

---

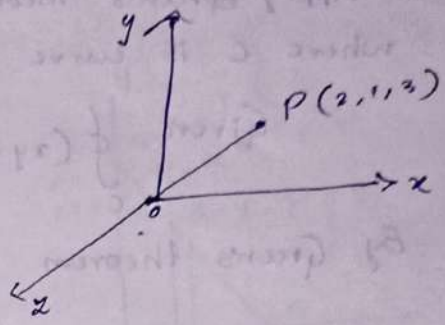
4. Compute the work done by the force $\bar{F} = 2x^2\bar{i} + (4xz - y)\bar{j} + 2z\bar{k}$ when it moves from along st line from point (0,0,0) to (2,1,3)

$\bar{F} = 2x^2\bar{i} + (4xz - y)\bar{j} + 2z\bar{k}$

$$\oint_C \bar{F}\cdot d\bar{r} = \oint_C (2x^2\bar{i} + (4xz - y)\bar{j} + 2z\bar{k})\cdot(i\,dx + j\,dy + k\,dz)$$

$= \oint_C 2x^2 dx + 4xz - y\,dy + 2z\,dz$

workdone by force $\bar{F}$, when moves a particle along straight line from $O(0,0,0)$ to $P(2,1,3)$



$$W = \oint_C \bar{F}\cdot d\bar{r}$$

$$= \oint_C 2(2t)^2 2\cdot dt + [4(2t)(3t) - t)]\,dt + 2(3t)3\,dt$$

St line $= \dfrac{x-x_1}{x_2-x_1} = \dfrac{y-y_1}{y_2-y_1} = \dfrac{z-z_1}{z_2-z_1}$

$= \dfrac{x-0}{2-0} = \dfrac{y-0}{1-0} = \dfrac{z-0}{3-0} = t$

$\therefore x = 2t \Rightarrow dx = 2\,dt$

$y = t \Rightarrow dy = 1\,dt$

$z = 3t \Rightarrow dz = 3\,dt$

$$W = \oint_C (16t^2 + 24t^2 - t + 18t)\,dt$$

$$= \int_{t=0}^{t=1} (40t^2 + 17t)\,dt$$

$$= \frac{40t^3}{3} + \frac{17t^2}{2}$$

$$= \frac{40}{3} + \frac{17}{2}$$

$$= \frac{40\times 2 + 17\times 3}{6} = \frac{80+51}{6} = \frac{131}{6}.$$

$\Rightarrow$ Green's theorem for plane (XY-plane)

Let $M(x,y)$, $N(x,y)$, $M_y = \left(\dfrac{\partial M}{\partial y}\right)$, $N_x = \left(\dfrac{\partial N}{\partial x}\right)$, be continuous differentiable function over the region E, in coordinate XY-plane, then

$$\boxed{\oint_C M(x,y)\,dx + N(x,y)\,dy = \iint_E \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy.}$$

$\downarrow$ short form

$$\boxed{\oint_C M\,dx + N\,dy = \iint_E \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx\,dy}$$

Session - 26

1. Apply Green's theorem to evaluate the integral $\oint_C (xy - y^2)dx - x^2 dy$ where C is curve bounded by $y = x$ & $y = x^2$.

Given $\oint_C (xy - y^2)dx - x^2 dy$

By Green's theorem $\oint_C M dx + N dy = \iint_G \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy$.

where $M = xy - y^2$, $N = -x^2$

$\frac{\partial M}{\partial y} = x - 2y$; $\frac{\partial N}{\partial x} = -2x$.

$= \oint_C (xy - y^2)dx - x^2 dy$

Session - 26

2. Apply Green's theorem, evaluate $\oint_C (3x - 8y^2)dx + (4y - 6xy)dy$. where C is boundary of region bounded by $x = 0, y = 0, x + y = 1$

Given $\oint_C (3x - 8y^2)dx + (4y - 6xy)dy$.

By Green's theorem for plane:

$$\oint_C M dx + N dy = \iint_S \left(\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}\right) dx dy.$$

where, $M = 3x - 8y^2$, $N = 4y - 6xy$

$\frac{\partial M}{\partial y} = -16y$ $\frac{\partial N}{\partial x} = -6y$

x Limits from 0 to 1

y Limits from 0 to $1 - x$

$= \oint$ ~~$-16y\,dx - 6y\,dy$~~

$$\oint_C (3x-8y^2)\,dx + (4y-6xy)\,dy = \iint_S -6y+16y \; dx\,dy$$

$$= \iint_S 10y \; dx\,dy.$$

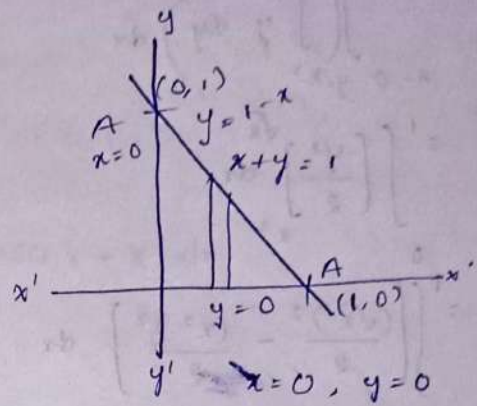
$$= \int_{x=0}^{1} \left(\int_{y=0}^{1-x} 10y \; dy \right) dx$$

$$= \int_0^1 \left[\frac{10y^2}{2} \right]_0^{1-x} dx$$

$$= \int_0^1 5(1-x)^2 \; dx = \int_0^1 5(1+x^2-2x)\,dx$$

$$= \left[5\left(x + \frac{x^3}{3} - \frac{2x^2}{2} \right) \right]_0^1 = \left[5x + \frac{5x^3}{3} - \frac{\cancel{10}^{5}x^2}{\cancel{2}} \right]_0^1$$

$$= \cancel{5} + \frac{5}{3} - \cancel{5} = 5/3 //.$$



$x=0,\ y=0$

$$\frac{x}{1} + \frac{y}{1} = 1$$

---

3. Apply Green's theorem. $\oint_C (3x^2-2y^2)\,dx + (4y-3xy)\,dy$ where C is bounded by curves $y=\sqrt{x}$, $y=x^2$.

$M = 3x^2-2y^2$ $\qquad$ $N = 4y-3xy$

$\frac{\partial M}{\partial y} = -4y$ $\qquad$ $\frac{\partial N}{\partial x} = -3y$

$$= \cancel{\iint} \iint_S \cancel{(-3y+4y)\,dx\,dy} \quad \iint_S -3y+4y \; dx\,dy$$

$$\cancel{\iint 4/4} \qquad = \iint_S y \; dx\,dy$$

$y=\sqrt{x}$

$x=\sqrt{y}$

$$\oint_C (3x^2-2y^2)\,dx + (4y-3xy)\,dy = \iint_S y \; dx\,dy$$

where S is region bounded by curves $y=\sqrt{x}$ & $y=x^2$

$y^2 = x$

Limits $y = x^2 \to \sqrt{x}$

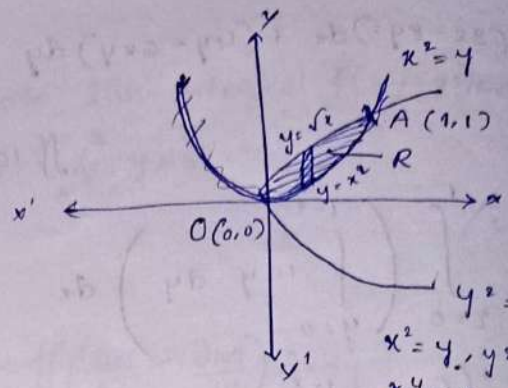
$x = 0 \to 1$

$$= \int_{x=0}^{1}\left(\int_{y=x^2}^{\sqrt{x}} y \, dy\right) dx$$

$$= \int_0^1 \left[\frac{y^2}{2}\right]_{x^2}^{\sqrt{x}} dx$$

$$= \int_0^1 \left[\frac{(\sqrt{x})^2}{2} - \frac{(x^2)^2}{2}\right] dx = \int_0^1 \frac{x}{2} - \frac{x^4}{2} \, dx$$

$$= \left[\frac{x^2}{4} - \frac{x^5}{10}\right]_0^1 = \frac{1}{4} - \frac{1}{10} = \frac{10-4}{40} = \frac{\cancel{6}\,3}{\cancel{40}\,20} = 3/20 \,//.$$



$x^2 = y, \; y^2 = x$

$x^4 = x \Rightarrow$

$x(x^3 - 1) = 0$

$x = 0, \; x = 1$

$y = 0, \; y = 1$

---

4. Apply Green's theorem to evaluate the integral over $\oint_C [y - \sin x]\,dx + \cos x \, dy$, where c is plane triangle enclosed by lines $y = 0$, $x = \pi/2$ & $y = \frac{2}{\pi} x$.

$M = y - \sin x \qquad N = \cos x$

$\frac{\partial M}{\partial y} = 1 \qquad \frac{\partial N}{\partial x} = -\sin x.$

$$\oint_C [y - \sin x]\,dx + \cos x \, dy = \iint_R -\sin x - 1 \, dx\,dy$$

Region is plane triangle enclosed by the lines

$y = 0, \; x = \pi/2$ & $y = (2/\pi)x$ — $y = mx$

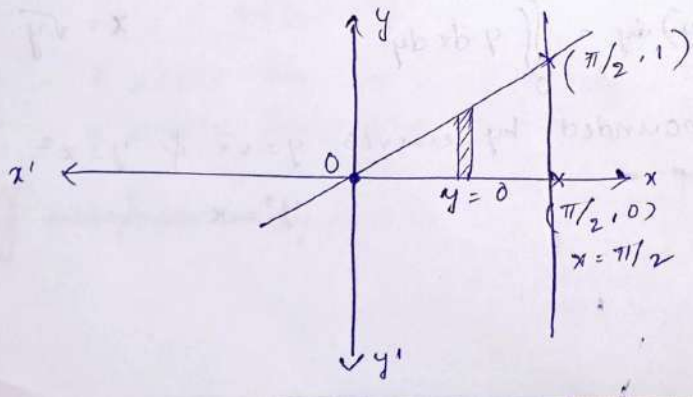
$y = 0$

$x = \pi/2$

$y = 2/\pi \; x$

$y = \cancel{2}/\cancel{\pi} \times \cancel{\pi}/\cancel{2}$

$y = 1$



$$= \int_{x=0}^{\pi/2} \left(\int_{y=0}^{(2/\pi)x} (-\sin x - 1)\, dy \right) dx$$

$$= \int_0^{\pi/2} \left[-y \sin x - y \right]_0^{2/\pi\, x} dx$$

$$= \int_0^{\pi/2} -\tfrac{2}{\pi} x \sin x - \tfrac{2}{\pi} x \, dx = -\frac{2}{\pi} \int_0^{\pi/2} x \sin x + x \, dx$$

$$= -\tfrac{2}{\pi} \int_0^{\pi/2} x \sin x - \tfrac{2}{\pi} \int_0^{\pi/2} +x \, dx$$

ILATE

$$\int f(x)\, g(x)$$

$$= f(x) \int g(x)\, dx - \int \left(f'(x) \cdot \int g(x)\, dx \right) dx$$

$$= -\tfrac{2}{\pi} \left[\left[x(-\cos x) - \int 1 \cdot -\cos x \, dx \right] \right]_0^{\pi/2} - \tfrac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi/2}$$

$$= -\tfrac{2}{\pi} \left[-x \cos x + \sin x \right]_0^{\pi/2} - \tfrac{2}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi/2}$$

$$= -\tfrac{2}{\pi} \left[\left(-\tfrac{\pi}{2} \cos \tfrac{\pi}{2} + \sin \tfrac{\pi}{2} \right) - (0 + 0) \right] - \tfrac{2}{\pi} \frac{(\pi/2)^2}{2}$$

$$= -\tfrac{2}{\pi} (1) - \tfrac{2}{\pi} \left(\frac{\pi^2}{8\,4} \right)$$

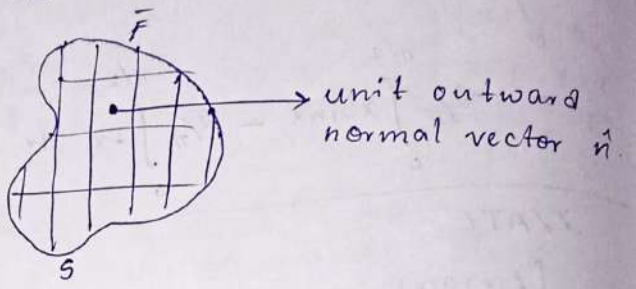
$$= -\left[\tfrac{2}{\pi} + \tfrac{\pi}{4} \right] //.$$

Session - 27

1. Surface integral
2. Stoke's theorem

1. Surface integral: Consider a continuous vector point function $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$. If $\hat{n}$ is a unit outward normal vector to the surface at any point, then surface integral of $\bar{F}$, over the given surface 's' is

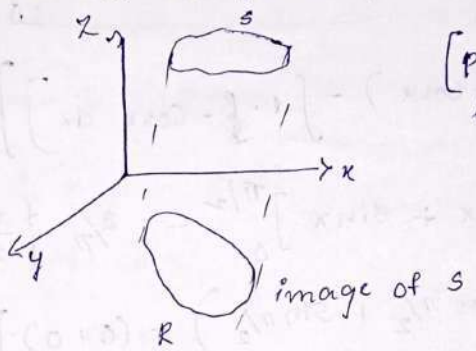
$$\oint_S \bar{F}\cdot\hat{n}\, ds$$



Projects of surface:-

(i) The projection of surface on XY-plane.

$$ds = \frac{dx\,dy}{|\bar{n}\cdot\bar{k}|}$$



[projection of XY-plane is Z ie $\bar{n} = \bar{k}$]

(ii) Projection on YZ - plane

$$ds = \frac{dy\,dz}{|\bar{n}\cdot i|}$$

(iii) Projection of surface of XZ - plane

$$ds = \frac{dx\,dz}{|\bar{n}\cdot j|}$$

2. Stoke's theorem (Relation btw Line & Surface integral)

If s be a open surface bounded by a closed curve 'C' & $\bar{F} = F_1\bar{i} + F_2\bar{j} + F_3\bar{k}$, be continuously differentiable over the

vector point function, then

$$\boxed{\oint_C \bar{F}\cdot d\bar{r} = \oint_S \text{Curl}\,\bar{F}\cdot N\, ds}$$

where N is unit external normal vector to surface 'S'

$$\boxed{N = \bar{k}\ ;\ ds = \frac{dx\,dy}{|\bar{k}\cdot\bar{k}|} = dx\,dy} \qquad \frac{dx\,dy}{|\bar{k}\cdot\bar{k}|} = \frac{dx\,dy}{|n\cdot\bar{k}|}$$

1. Apply stoke's theorem to evaluate $\oint_C \bar{F}\cdot d\bar{r}$, where $\bar{F} = y^2 i + x^2 j - (x+z)\bar{k}$ & C is boundary of triangle with vertices at $(0,0,0)$, $(1,0,0)$ & $(1,1,0)$

Given vector point function,

$$\bar{F} = y^2\bar{i} + x^2\bar{j} - (x+z)\bar{k}$$

By stokes theorem. $\oint_C \bar{F}\cdot dr = \oint_S \text{Curl}\,\bar{F}\cdot N\,ds$

Now, $\text{Curl}\,\bar{F} = \nabla\times\bar{F}$

$$\begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^2 & x^2 & -(x+z) \end{vmatrix}$$

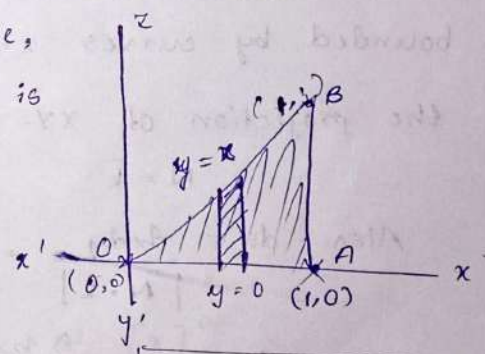
$$= \bar{i}([0]-(0]) - \bar{j}(-1-0) + \bar{k}(2x-2y)$$

$$= +\bar{j} + \bar{k}(2x-2y)$$

S is surface of triangle $(0,0,0)$, $(1,0,0)$, $(1,1,0)$.

Since triangle region lies on XY-plane, the projection of XY-plane is Z-axis is

i.e $\boxed{N = \bar{k}}$



$\boxed{z = 0 \Rightarrow XY\text{-plane}}$

Also, $ds = \frac{dx\,dy}{|N\cdot\bar{k}|} = \frac{dx\,dy}{|\bar{k}\cdot\bar{k}|} = dx\,dy$.

$$= \oint_C \bar{F}\cdot d\bar{r} = \iint_R [\bar{j} + (2x-2y)\bar{k}]\cdot\bar{k}\cdot dx\,dy$$

$$= \iint_R (2x-2y)\,dx\,dy$$

$$= \int_{x=0}^{1}\left(\int_{y=0}^{x}(2x-2y)\,dy\right)dx$$

$$= \int_{0}^{1}\left[2xy - \frac{2y^2}{2}\right]_0^x dx$$

$$= \int_0^1 2x^2 - \frac{\cancel{2}x^2}{\cancel{2}}\,dx \Rightarrow \int_0^1 x^2\,dx$$

$$= \left[\frac{x^3}{3}\right]_0^1 = 1/3 \,//.$$

---

2. Apply Stokes theorem, for vector field $F = (x^2 - y^2)i + 2xy\,j$ integrate around rectangle in plane $z = 0$ & bounded by curves $x = 0,\ y = 0,\ x = a,\ y = b$

$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ (x^2 - y^2) & 2xy & 0 \end{vmatrix}$$

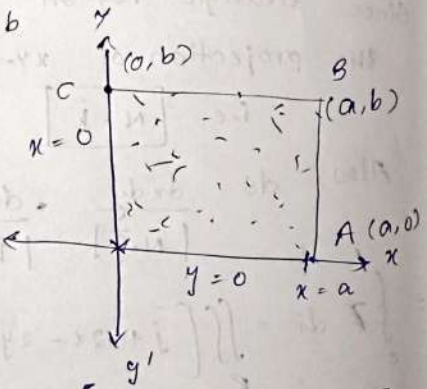
$$= \bar{i}(0-0) - \bar{j}(0-0) + \bar{k}(2y + 2y)$$

$$= 4y\bar{k}$$

$$\Rightarrow \oint_C F \cdot d\bar{r}$$

S is a surface around rectangle in plane $z = 0$ bounded by curves $x = 0,\ y = 0,\ x = a,\ y = b$

the projection of XY- plane



[Z = 0 XY - plane]

$$N = \bar{k}$$

Also $ds = \dfrac{dx\,dy}{|N \cdot \bar{k}|} = \dfrac{dx\,dy}{|\bar{k} \cdot \bar{k}|} = dx\,dy\,x'$

$$= \oint_C F \cdot dr = \int_{x=0}^{a}\int_{y=0}^{b} 4y\bar{k} \cdot \bar{k}\,dx\,dy$$

$$= \int_{x=0}^{a}\left(\int_{y=0}^{b} 4y\right)dy\Big)\,dx$$

$$= \int_0^a \left[\frac{4y^2}{2}\right]_0^b dx \qquad = \int_0^a 2b^2\, dx$$

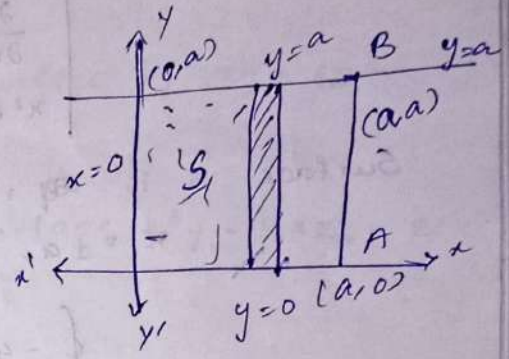
$$= [2b^2 x]_0^a = 2b^2 a \Rightarrow 2ab^2 \,//.$$

3. Apply stoke's theorem for vector field $\bar{F} = x^2 i + xy j$ integrate around square

$x = 0, y = 0, \; x = a, \; y = a$

$\bar{F} = x^2 i + xy j$

Square in plane $z = 0$



$$\text{curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2 & xy & 0 \end{vmatrix}$$

$$= i(0-0) - j(0-0) + k(y-0)$$

$$= yk.$$

$$= \oint_c F \cdot d\bar{r}$$

$x = 0, \; y = 0, \; x = a, \; y = a$

the projection of XY-plane

$N = \bar{k}$

Also $ds = \dfrac{dx\,dy}{|N \cdot \bar{k}|} = \dfrac{\partial x\, \partial y}{|\bar{k} \cdot \bar{k}|} = dx\,dy$

$$= \oint_c \bar{F} \cdot dr = \int_{x=0}^{a} \int_{y=0}^{a} y\bar{k} \cdot \bar{k}\, dx\, dy$$

$$= \int_{x=0}^{a} \left(\int_{y=0}^{a} y\, dy \right) dx$$

$$= \int_0^a \left[\frac{y^2}{2}\right]_0^a dx = \int_0^a \frac{a^2}{2}\, dx \Rightarrow a^2/2\, [x]_0^a$$

$$= a^3/2.$$

4. Apply stokes theorem to evaluate $\oint_c \bar{F}\cdot d\bar{r}$ where $\bar{F} = (x^2+y^2)\bar{i} - 2xy\bar{j}$ where c is the boundary of the rectangle bounded by lines $x = \pm a,\ y = 0,\ y = b$.

$$\bar{F} = (x^2+y^2)i - 2xy\,j$$

By stoke's theorem $\oint_c F\cdot d\bar{r} = \oint_c \text{Curl}\,\bar{F}\cdot N\,ds$.

$$\text{curl}\,\bar{F} = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2+y^2 & -2xy & 0 \end{vmatrix} = \bar{i}(0-0) - \bar{j}(0-0) + \bar{k}(-2y-2y)$$

$$= -4y\bar{k}$$

Surface S is rectangle in plane $z=0$ i.e, xy-plane.

$x = \pm a,\ y = 0,\ y = b$.

$$\oint_c -4y\bar{k}\cdot\bar{k}\,ds \Rightarrow \iint_S -4y\bar{k}\bar{k}\,dx\,dy$$

$$= \int_{x=-a}^{a}\left(\int_{y=0}^{b} -4y\,dy\right)dx$$

$$= \int_{x=-a}^{a}\left[\frac{-4y^2}{2}\right]_0^b dx = \int_{x=-a}^{a} -2b^2\,dx$$

$$= \left[-2b^2x\right]_{-a}^{a}$$

$$= -2b^2a - (-2b^2(-a))$$

$$= -4ab^2 //.$$

Tutorial - 10

1. Find gradient of the function $f = x^3y^2 - y^3z - 7(z^3+x)$ at the point $(1,2,1)$

2. Find divergence & curl of $\bar{f} = (2xyz)\bar{i} - (x^2y)\bar{j} - 3y^2z\bar{k}$ at $(1,1,1)$

3. Compute directional derivative by $f = xy^2 - 2y^3z$ in direction of vector $\bar{i} + 2\bar{j} + 2\bar{k}$ at point $(2,-1,1)$

4. Identify angle between normal to surface by $xy = 2z^2$ at points $(1,2,3)$ & $(2,2,2)$

5. Find unit normal vector to the surface $x^3y - 4xz = 2$ at point $(1,-1,3)$

1.

$$\text{Grad } \bar{f} = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}$$

$$= \frac{\partial}{\partial x}\left(x^3y^2 - y^3z - 7(z^3+x)\right) + \frac{\partial}{\partial y}\left(x^3y^2 - y^3z - 7(z^3+x)\right) + \frac{\partial}{\partial z}\left(x^3y^2 - y^3z - 7(z^3+x)\right)$$

$$= 3x^2y^2 - 0 - 7 + x^3y - 3y^2z - 0 + 0 - y^3 - 21z^2$$

$$= (3x^2y^2 - 7)i + (x^3y - 3y^2z)j - y^3 - 21z^2)k$$

$(1,2,1)$ $$= 3(1)^2(2)^2 - 7 + (1)^3(2) - 3(2)^2(1) - (2)^3 - 21(1)^2 - 7.$$

$$= \cancel{12 - 2 - 12}$$

$$\Rightarrow (12 - 7)i + [(1)^3(2) - 3(2)^2(1)]j + (-y^3 - 21z^2)k$$

$$= 5i - 8j - 29k.$$

$$\begin{array}{r} 21 \\ 8 \\ \hline 21 \end{array}$$

2. $f = (2xyz)i - (x^2y)\bar{j} - 3y^2z\bar{k}$ at $(1,1,1)$

$$\boxed{\begin{array}{l} \text{div} f = -2 \\ \text{curl} f = -6i + 2j \end{array}}$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & -x^2y & -3y^2z \end{vmatrix} = i(-6yz - 0) - j(0 - 2xy) + k(-2xy - 2xz)$$

$$= -6yzi + 2xyj + k(-2xy - 2xz)$$

$$= -6\bar{i} + 2\bar{j} - 4\bar{k}$$

$$\text{curl} f = -6\bar{i} + 2\bar{j} - 4\bar{k}$$

$$\text{div} f = \frac{\partial f_1}{\partial x} + \frac{\partial f_2}{\partial y} + \frac{\partial f_3}{\partial z}$$

$$= [2yz - 2x^2y - 0] + [2xz - x^2 - 6yz] + [2xy - 0 - 3y^2]$$

$$= 2 - 2 + 2 - 1 - 6 + 2 = 3$$

$$= 2yz - x^2 - 3y^2$$

$$= 2 - 1 - 3 = -2$$

---

3. $D.D = \nabla f \cdot \frac{\bar{a}}{|\bar{a}|}$

$$f = xy^2 - 2y^3z$$

$a \Rightarrow i + 2\bar{j} + 2\bar{k}$ at $(2,-1,1)$

$$\bar{a} = i + 2j + 2k$$

$$|\bar{a}| = \sqrt{1+4+4} = 3$$

$1 - 20 + 4$

$$\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}$$

$$= y^2 i + (2xy - 6y^2z)j + (-2y^3)k$$

$$= (-1)^2 i + [2(2)(-1) - 6(-1)^2(1)]j + [-2(-1)^3]k$$

$$= \frac{(i - 10j + 2k) \cdot (i + 2j + 2k)}{3} = \frac{1 - 20 + 4}{3} = -\frac{15}{3} = -5$$

4. $\cos\theta = \dfrac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|}$

$xy = 2z^2$

$f = xy - 2z^2$

n_1 at $(1,2,3)$ & n_2, at $(2,2,2)$

$\bar{n}_1 = \nabla f$ at $(1,2,3) = i\dfrac{\partial f}{\partial x} + j\dfrac{\partial f}{\partial y} + k\dfrac{\partial f}{\partial z}$

$= (y)i + (x)j + (-4z)k$

$= 2i + j - 12k.$

$\bar{n}_2 = \nabla f$ at $(2,2,2) = i\dfrac{\partial f}{\partial x} + j\dfrac{\partial f}{\partial y} + k\dfrac{\partial f}{\partial z}$

$= (y)i + (x)j + (-4z)k$

$= 2i + 2j - 8k$

$\cos\theta = \dfrac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|} = \dfrac{4+2-96}{\sqrt{4+1+144}\sqrt{4+4+64}}$

$= \dfrac{-90}{\sqrt{149}\sqrt{72}}$

---

5. $x^3y - 4xz = 2$ at $(1,-1,3)$

$\hat{n} = \dfrac{\nabla f}{|\nabla f|}$

$\nabla f = \dfrac{\partial f}{\partial x} + \dfrac{\partial f}{\partial y} + \dfrac{\partial f}{\partial z}$

$= (3x^2y - 4z)i + (x^3)j + (-4x)k$

$= [3(1)^2(-1) - 4(3)]i + (1)^3 j - 4k$

$= -15i + j - 4k$

$|\nabla f| = \sqrt{225 + 1 + 16} \Rightarrow \sqrt{242} = \sqrt{2 \times 121}$

$\Rightarrow 11\sqrt{2}$

$\hat{n} = \dfrac{-15i + j - 4k}{11\sqrt{2}}.$

$\begin{array}{r} 3 \\ 12 \\ \hline 15 \times 16 \\ \hline 75 \\ 15\ 2 \\ \hline 225 \\ 116 \\ \hline 242 \end{array}$

Tutorial - 11

1. A vector field $\bar{F} = (2y+3)\bar{i} - 2xz\,j - 3(yz - x)\bar{k}$

Evaluate $\oint_C \bar{F}\cdot d\bar{r}$ along path c. is $x = 2t$, $y = t$, $z = t^3$ from $t = 0$ to $t = 1$

2. If $\bar{F} = (x^2+y^2)i - 2xy\,j$, evaluate $\oint_C \bar{F}\cdot d\bar{r}$ where c is rectangle in xy-plane bounded by $x=0, x=a, y=0, y=b$ $\boxed{-2a^2b}$

3. Find $\oint_C \bar{F}\cdot d\bar{r}$ where $\bar{F} = x^2y^2 i + y\,j$ & the curve $y^2 = 4x$ in xy-plane from $(0,0)$ to $(4,4)$.

1. $\oint_C \bar{F}\cdot d\bar{r}$

$$= \oint_C (2y+3)i - (2xz)j - 3(yz-x)k \cdot (i\,dx + j\,dy + k\,dz)$$

$$= \oint_C 2y+3\cdot dx - 2xz\cdot dy - 3(yz-x)\cdot dz$$

$x = 2t \Rightarrow dx = 2dt$

$y = t \Rightarrow dy = dt$

$z = t^3 \Rightarrow dz = 3t^2 dt$

$$= [2(t)+3]\cdot 2dt - 2(2t)(t^3)dt - [3(t(t^3) - 2t)3t^2]dt$$

$$= \int_0^1 4t + 6 - 4t^4 - 9t^6 + 18\cdot t^3 \, dt$$

$$= \left[\frac{\cancel{4}^{2}t^2}{\cancel{2}} - 6t - \frac{4t^5}{5} - \frac{9t^7}{7} + \frac{\cancel{18}^{9}t^4}{\cancel{4}_{2}}\right]_0^1$$

$$= 2 - 6 - 4/5 - 9/7 + 9/2$$

$$= -4 - 4/5 - 9/7 + 9/2$$

$$= -\frac{9\times 2 + 9\times 7}{14} = -\frac{18+63}{14} = -\frac{20-4}{5} + \frac{-24}{5}$$

35×2
70

-24
$-\frac{15}{4}$

$\frac{63}{14}$
$\frac{45}{14}$

CO-3 Revision problems

(1) Apply $\beta-\Gamma$ functions, evaluate the integral $\int_0^{\pi/2} \sin^3\theta \cos^7\theta \, d\theta$

(2) Evaluate integral $\iint xy \, dx \, dy$ in the positive quadrant for which $x+y \le 1$.

(3) Evaluate the integral $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2) \, dy \, dx$ by changing into polar coordinates.

1. $2m-1 = 3$ $\quad$ $2n-1 = 7$

$2m = 4$ $\quad$ $2n = 8$

$\boxed{m = 2}$ $\quad$ $\boxed{n = 4}$

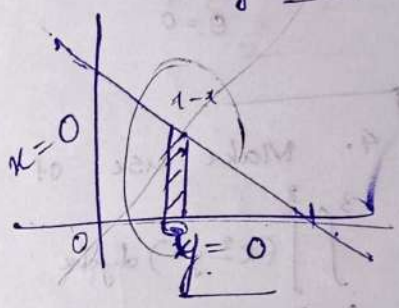
$$= 2 \cdot \frac{1}{2} \int_0^{\pi/2} \sin^{2(2)-1}\theta \cos^{2(4)-1}\theta \, d\theta = \beta(2,4)$$

$$\Rightarrow \frac{1}{2} \cdot \frac{\Gamma(2)\Gamma(4)}{\Gamma(6)} = \frac{1}{2} \cdot \frac{1! \times 3!}{5!}$$

$$= \frac{\cancel{3 \times 2}}{2 \times 5 \times \cancel{4} \times \cancel{3 \times 2} \times 1} = \frac{1}{40}.$$

$x+y=1$

$y = 1-x$.



$\iint xy \, dx \, dy$

$x+y \le 1$.

$$\int_{x=0}^{1} \left(\int_{y=0}^{1-x} xy \, dy \right) dx$$

$$\int_0^1 \left[\frac{xy^2}{2} \right]_0^{1-x} dx \Rightarrow \int_0^1 \frac{x(1-x)^2}{2} dx = \int_0^1 \cancel{\frac{x-x^2}{2}} \, dx$$

$$= \frac{1}{2} \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 \Rightarrow \frac{1}{2} \left(\frac{1}{2} - \frac{1}{3} \right)$$

$$= \frac{1}{2} \left(\frac{3-2}{6} \right)$$

$$= \int_0^1 \frac{x(1+x^2-2x)}{2} dx = \frac{1}{2} \int_0^1 x + x^3 - 2x^2 = \frac{1}{2} \left[\frac{x^2}{2} + \frac{x^4}{4} - \frac{2x^3}{3} \right]$$

$$= \frac{1}{2} \left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3} \right]$$

$$= \frac{1}{2}\left[\frac{1}{2} + \frac{1}{4} - \frac{2}{3}\right]$$

$$= \frac{1}{2}\left(\frac{6+3-8}{12}\right) = \frac{1}{24}.$$

$2/3 \approx 1/2$

$\frac{9-8}{12}$

$$\begin{array}{c|c} 2 & 2,4,3 \\ 2 & 1,2,3 \\ 3 & 1,1,3 \\ & 1,1,1 \end{array}$$

$4 \times 3 = 12$

3. $$\int_{x=0}^{1}\int_{y=0}^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx$$

$x = r\cos\theta,\ y = r\sin\theta$

$dx\,dy = r\,dr\,d\theta$

$$\Rightarrow \int_{\theta=0}^{\pi/2}\int_{r=0}^{1} r^2 \cdot r\,dr\,d\theta$$

$y = 0$ to $y = \sqrt{1-x^2}$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

$x = 0$ to $x = 1$

$$= \int_{\theta=0}^{\pi/2}\left[\frac{r^4}{4}\right]_0^1 dr\,d\theta \Rightarrow \frac{1}{4}\left[\theta\right]_0^{\pi/2}$$

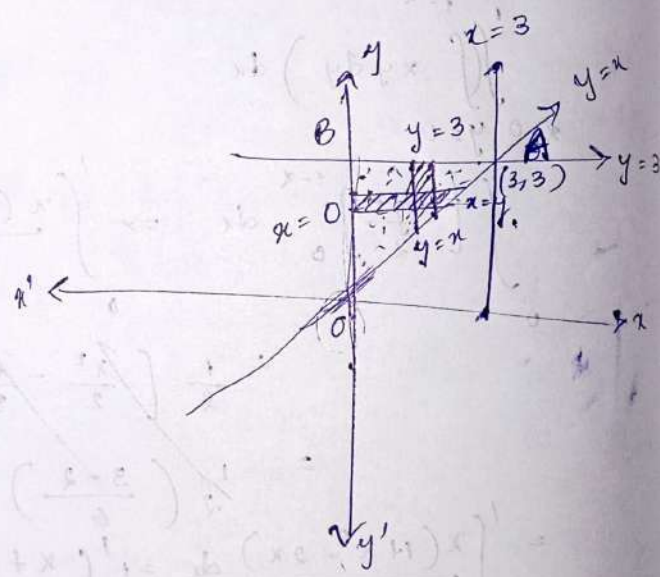
$$= \pi/8.$$

4. Make use of change the order of integration evaluate the $\int_0^3\int_x^3 (x^2-y^2)\,dy\,dx$

Given $$\int_{x=0}^{3}\int_{y=x}^{3} (x^2-y^2)\,dy\,dx.$$

$y = x$ to $y = 3$

$x = 0$ to $x = 3$



By changing the order of integration $$\int_{y=0}^{3}\left(\int_{x=0}^{y} (x^2-y^2)\,dx\right)dy$$

$$\int_0^3 \left[\frac{x^3}{3} - y^2 x\right]_0^y \, dy$$

$$\int_0^3 \frac{y^3}{3} - y^3 \, dy \Rightarrow \int_0^3 \frac{1}{3} \cdot \frac{y^4}{4} - \frac{y^4}{4}$$

$$\int_0^3 \left[\frac{y^4}{12} - \frac{y^4}{4}\right] \Rightarrow \frac{(3)^4}{12} - \frac{(3)^4 \times 3}{4}$$

$$\frac{(3)^4}{12} - \frac{(3)^4}{4} = \frac{81 - 243}{12} = \frac{-162}{12}$$

$$= -27/2$$

$3 \times 3 \times 3 \times 3$
$= 9 \times 9 = 81$

81×3
243

81
162

81
252

24
8
16

17
18
24

---

5. Evaluate integral $\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \, dx \, dy \, dz$.

$$\int_0^1 \int_0^1 \int_0^1 e^{x+y+z} \, dx \, dy \, dz$$

$$\int_{x=0}^1 \int_{y=0}^1 \int_{z=0}^1 e^x \cdot e^y \cdot e^z \, dz \, dy \, dx$$

$$\int_{x=0}^1 \int_{y=0}^1 [e^x]_0^1 \, e^y \cdot e^z \, dy \, dz$$

$$= \int_{x=0}^1 \int_{y=0}^1 [e^1 - e^0] \, dy \, dz. = \int_{x=0}^1 \left[\int_{y=0}^1 (e-1) \cdot e^y \, dy\right] dz$$

$$(e-1) \int_{z=0}^1 \int_{y=0}^1 e^y \, dy \cdot e^z \, dz$$

$$= (e-1)^3$$

e^{2x}

$= \frac{e^{2x} \times 2}{2}$

6. $$\int_{x=0}^{a}\int_{y=0}^{a}\int_{z=0}^{a}(xy+yz+zx)\,dx\,dy\,dz.$$

$$\int_{x=0}^{a}\int_{y=0}^{a}\int_{z=0}^{a}(xy+yz+zx)\,dz\,dy\,dx$$

$$\int_{x=0}^{a}\int_{y=0}^{a}\left[xyz+\frac{yz^2}{2}+\frac{z^2x}{2}\right]_0^a dy\,dx$$

$$=\int_{x=0}^{a}\int_{y=0}^{a}\left[xya+\frac{ya^2}{2}+\frac{a^2x}{2}\right]dy\,dx$$

$$=\int_{x=0}^{a}\left[\frac{xy^2a}{2}+\frac{y^2a^2}{4}+\frac{a^2xy}{2}\right]_0^a dx$$

$$=\int_{x=0}^{a}\frac{x(a)^2a}{2}+\frac{(a)^2a^2}{4}+\frac{a^2x(a)}{2}\,dx$$

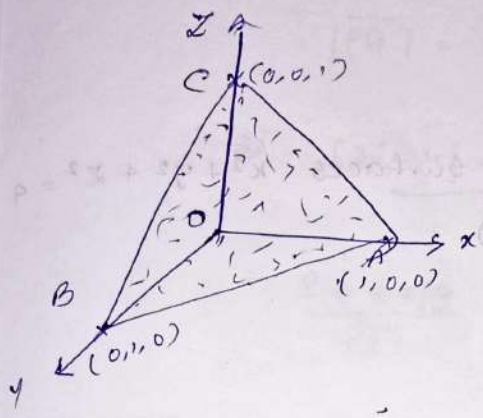
$$=\left[\frac{x^2a^3}{2\times 2}+\frac{a^4x}{4}+\frac{a^3x^2}{2\times 2}\right]_0^a$$

$$=\frac{a^2a^3}{4}+\frac{a^4a}{4}+\frac{a^3a^2}{4}$$

$$=\frac{a^5+\cancel{a^5}+\cancel{a^5}}{4}=\frac{3a^5}{4}$$

7. Evaluate volume of tetrahedron bounded by planes $x=0,\ y=0,\ z=0;\ x+y+z=1$.

$$\frac{x}{1}+\frac{y}{1}+\frac{z}{1}=1$$



OABC is tetrahedron

Volume of tetrahedron

$$= \iiint_V dx\,dy\,dz.$$

$$= \int_{x=0}^{1}\left(\int_{y=0}^{1-x}\left(\int_{z=0}^{1-x-y} dz\right) dy\right) dx$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x}[z]_0^{1-x-y}\,dy\,dx$$

$$= \int_{x=0}^{1}\int_{y=0}^{1-x} 1-x-y\,dy\,dx \Rightarrow \int_{x=0}^{1}\left[y-yx-\frac{y^2}{2}\right]_0^{1-x} dx$$

$$= \int_{x=0}^{1}(1-x)-(1-x)x-\frac{(1-x)^2}{2}\,dx$$

$$= \int_{x=0}^{1} 1-x-x+x^2-\frac{1-x^2+2x}{2}\,dx$$

$$= \int_{x=0}^{1}\frac{2-2x-\cancel{2x}+2x^2-1-x^2+\cancel{2x}}{2} = \int_{x=0}^{1}\frac{1-x^2-2x}{2}$$

$$\frac{1}{2}\left[x-\frac{x^3}{3}-\frac{\cancel{2}x^2}{\cancel{2}}\right]_0^1 \Rightarrow \frac{1}{2}\left(1-\frac{1}{3}-1\right) = -\frac{1}{6}.$$

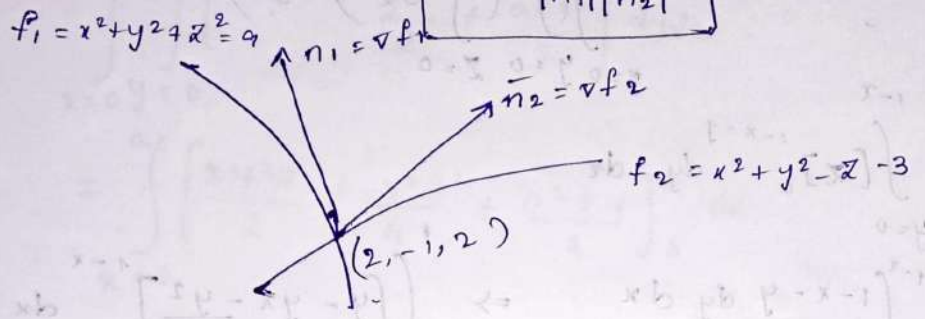
CO-4 Revision - Problems

1. Find the directional derivative of $f = x^2 y^2 + 2z^2$ at point $P(1,2,3)$ in direction of line PQ where Q is point $(5,0,4)$

$$\boxed{D.D = \nabla f \cdot \frac{\overline{PQ}}{|\overline{PQ}|}}$$

2. Identify the angle between the surfaces $x^2+y^2+z^2=9$ & $x^2+y^2-z=3$ at point $(2,-1,2)$

$$\boxed{\cos\theta = \frac{\bar{n}_1 \cdot \bar{n}_2}{|\bar{n}_1||\bar{n}_2|}}$$



3. Obtain divergence & curl of $\bar{f} = (xyz)\bar{i} - 3x^2 y j + y^2 z \bar{k}$ at point $(1,2,1)$

1. $f = x^2 - y^2 + 2z^2$

$$\nabla f = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z}$$

$$= (2x - y^2 x + 2z^2 x)i + (-2y + 2z^2 y)j + (4z - y^2 z + 4z)k$$

$$= (2 - (2)^2(1) + 2(1)^2(1))i + ((2)(1)^2 - 2(2) - 2(1)^2(2))j + ((1)^2(1) - (2)^2(1) + 4(1))k$$

$$= (2 - 4 + 2)i + (2 - 4 - 4)j + (1 - 4 + 4)k$$

$$= -6j + k.$$

$$\begin{array}{r} -8 \\ +2 \\ \hline -6 \end{array}$$

$\nabla f = 2x - 2y + 4z$

$= 2xi - 2yj + 4zk = 2i - 4j + 12k$

$\overrightarrow{PQ} = OQ - OP = 5i + 4k - i - 2j - 3k$

$= 4i - 2j + k.$

$|\overrightarrow{PQ}| = \sqrt{16+4+1} = \sqrt{21}.$

$$\frac{(2i - 4j + 12k)(4i - 2j + k)}{\sqrt{21}}$$

$$= \frac{8 + 8 + 12}{\sqrt{21}} = \frac{28}{\sqrt{21}} = \frac{28}{\sqrt{12}}.$$

---

2. $f_1 = x^2 + y^2 + z^2 = 9$

$2xi + 2yj + 2zk \Rightarrow 4i - 2j + 4k$

$f_2 = x^2 + y^2 - z = 3$

$= 2xi + 2yj - k \Rightarrow 4i - 2j - k.$

$$= \frac{(4i - 2j + 4k) \cdot (4i - 2j - k)}{\sqrt{16+4+16}\ \sqrt{16+4+1}}$$

$$= \frac{16 + 4 - 4}{\sqrt{36}\ \sqrt{21}} = \frac{8}{3\sqrt{21}} = \frac{8}{3\sqrt{21}}.$$

$$\theta = \cos^{-1}\left(\frac{8}{3\sqrt{21}}\right).$$